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Deep learning-assisted arrhythmia classification using 2-D ECG spectrograms

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Abstract

This article studies modern classification techniques in ECG signals through the transfer learning approach with CNN (Convolutional Neural Network). The proposed pre-trained network combines an Imagenet with huge labeled image datasets and a separate network composed of fully connected layers. This method uses the CWT (Continuous Wavelet Transform) to construct a time-frequency visualization of ECG signals, which are subsequently transformed into RGB images. The developed images are plugged into a pre-trained CNN to retrieve the desired features. We next employ supervised learning to train the neural network on the ECG labeled data using CNN features. To train a Deep Neural Network, three sets of PhysioNet databases are used: MIT-BIH (ARR) Arrhythmia, NSR (Normal Sinus Rhythm), and BIDMC CHF (Congestive Heart Failure). The classification Accuracy, Sensitivity, Specificity, F1-score, Precision, and Detection Error Rate of the CNN classifier are compared to AlexNet, GoogleNet, Vgg16, and SqueezeNet pre-trained networks. Among all these networks, SqueezeNet provides an Acc of 98.7%, Se of 99.1%, Sp of 99.20%, F1-score of 98.33%, Precision of 98.67%, and DER of 0.89%. For further investigation, the technique suggested can be implemented in addition to Bi-LSTM on some real ECG data.

Keywords: Electrocardiogram, Convolutional Neural Networks, Classification, Deep learning

1 Introduction

The heartbeat, a fundamental physiological signal, is a crucial indicator of cardiac activity, reflecting various anomalies influenced by age and lifestyle. Deep learning, pivotal in modern computer-based solutions, facilitates automated feature detection from input data. It enables information extraction, prediction, and intelligent decision-making, leveraging training datasets. Heart irregularities are increasing, posing global health concerns. ECG signals remain key for detecting and assessing heart abnormalities. Sedentary lifestyles contribute to rising heart and cancer issues. Xu [1] describes an IoT-assisted ECG tracking system that ensures secure data transfer for healthcare. It employs IoT devices for remote ECG data collection, securely transmitting it to healthcare providers.

The authors [2, 3] addressed an efficient classification method using Convolutional Neural Networks (CNN) by monitoring healthy heart activity via ECG analyzing P, QRS,

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T waves, and QT intervals. Shaker et al. [4] introduced a new data augmentation method with generative adversarial network (GAN) to balance the dataset as well as detect heart rate-affecting conditions. The authors applied two deep CNN methodologies from end to end and two-stage model. This method integrates feature extraction, reduction, and classification of heart beats. Hsu and Liu [5] implemented a new strategy to fold ECG signals into fingerprint format, which could have implications for efficient storage, transmission, and analysis of ECG data, potentially advancing the diagnostics of cardiovascular and healthcare technology.

Wasimuddin et al. [6] presented a review on ECG signal analysis using machine learning approaches like neural networks and support vector machines (SVMs). Improper signals can lead to stroke or heart failure. NSR signals a healthy heart, while CHF indicates compromised functionality due to inadequate circulation. To overcome this problem, Elgendi and Menon [7] presented a novel deep learning (DL) algorithm by combining CNN and bi-directional long short-term memory (Bi-LSTM). This algorithm not only increases the classification accuracy but also reduces the training time. Gao [8] presented a study on DL models such as CNN, RNN, LSTM, DBN, etc. for the identification of irregular heartbeats. Many researchers have recommended a variety of DL models to increase the accuracy of multiple learning tasks, including multilayer perception (MLP), CNN, RNN, long short-term memory (LSTM), and deep belief network (DBN) [9]. Yildirim et al. [10] projected a novel algorithm for the classification of cardiac disorders, which enhances the signal quality. A deep CNN is employed to obtain the features and then fed into an LSTM network, which process the sequential data. Incorporating deep coded features with LSTM networks creates an effective strategy for arrhythmia classification, possibly enhancing the accuracy and efficacy of automated ECG monitoring in clinical environments.

A CNN method with improved accuracy for automatically detecting regular and myocardial infarction ECG beats in both noisy and noise-free conditions is investigated by [11]. This method attained an average accuracy of 93.53% and 95.22% for ECG beats with and without noise reduction, respectively. Furthermore, this study did not involve any feature extraction or selection. As a result, this method can recognize unfamiliar ECG signals with high accuracy even in noisy conditions. Malleswari et al. [12] implemented a pre-trained neural network to analyze cardiac severity classification. The classification of cardiac severity is accomplished through the utilization of existing models. In order to enhance the efficacy and accuracy of cardiac diagnostics, pre-trained networks are implemented.

Ting [13] developed an algorithm for the detection of AF (atrial fibrillation) using ECG signals. This technique uses machine learning to accurately identify AF with high accuracy and reliability. Hannun [14] introduced a deep neural network to train an ambulatory ECG records, employing CNN for the extraction and classification of features. A spectrogram is created by converting a one-dimensional (1-D) ECG waveform into two-dimensional (2-D) signals using short time Fourier transform (STFT), which is then used as a CNN classifier. Amin et al. [15] study proposes a 2D CNN model for classifying ECG signals into eight categories, including arrhythmias. Initially, 2D spectrograms are obtained by applying STFT on 1D ECG signals. The presented CNN architecture consists of four convolutional and pooling layers for feature extraction and classification,

which provides high sensitivity and specificity, indicating its effectiveness in arrhythmia classification.

Hong [16] investigated the usage of deep learning in ECG data processing with an emphasis on CNN and RNN for the detection and classification of arrhythmia. It addresses new methodologies such as transfer learning and federated learning. Gaddam [17] presented a transferred DL algorithm to classify the ECG signal into four classes such as ARR, NSR, Malignant Ventricular Ectopy (MVE), and CHF. In this method, 2D scalogram images are trained using CNN with pre-trained AlexNet with an accuracy of 95.67%. To enhance the classification accuracy with CNN, Jia et al. [18] fused the structure and tempo of heartbeats to get a 2D information vector; then, it was processed using a flexible learning rate and weighted dropout techniques. Malleswari et al. [19] presented a novel algorithm that optimizes the detection of QRS complexes in ECG signals by integrating DWT and EMD. In experiments conducted on a variety of ECG datasets, this hybrid approach outperforms standard techniques concerning reliability as well as accuracy.

Rahman et al. [20] implemented a deep CNN with transfer learning strategy to classify heartbeats and assessed it on PhysioNet's MIT-BIH and PTB Diagnostics datasets. Novel deep learning algorithms are developed based on CNN with bidirectional LSTM showing more accuracy and less training time [21]. Panda et al. [22] introduced a fixed frequency range empirical wavelet transform (FFREWT) deep CNN for the detection of SVCA (Shockable Ventricular Cardiac Arrhythmias), and the obtained average accuracy is 97.592%. To enhance the classification accuracy, Kanani and Padole [23] offer a modified DL architecture with stability. To extract the QRS complex, a DTCWT (dual-tree complex wavelet transform) feature extraction method is applied, and its performance is compared to that of the DWT using a multi-layer BPNN by [24]. A mobile device-optimized deep neural network (DNN) for ECG signal classification is introduced by [25]. The DNN correctly classifies signals with minimum processing cost on ECG datasets, allowing mobile cardiac monitoring in real time.

In this work, we transformed 1D ECG signals into 2D images and used CWT to extract time-frequency components, which were then fed to pre-trained AlexNet, GoogleNet, Vgg16, and SqueezeNet CNN classifiers. Finally, their performance measures are compared with existing methods and this paper uses the MIT-BIT arrhythmia database as the data for the experiments in the following chapters.

2 Materials and methods

The projected system involves three stages: generating ECG signal time-frequency components, converting 1D signals into 2D images, and using pre-trained networks for transfer learning and CNN classification. The CWT-CNN classifier is compared with Alexnet, GoogleNet, Vgg16, and Squeeze net models in terms of Acc, Se, Sp, Precision, F1-score, and DER. The proposed pre-trained CNN architecture is shown in Fig. 1, which consists of ten 2D convolutional layers and each layer is followed by a pooling layer. To provide the ultimate classification, a softmax layer of three neurons is employed as the output. A fully connected layer works across the last layer of pooling and the output layer to represent the attributes learnt by the CNN algorithm.

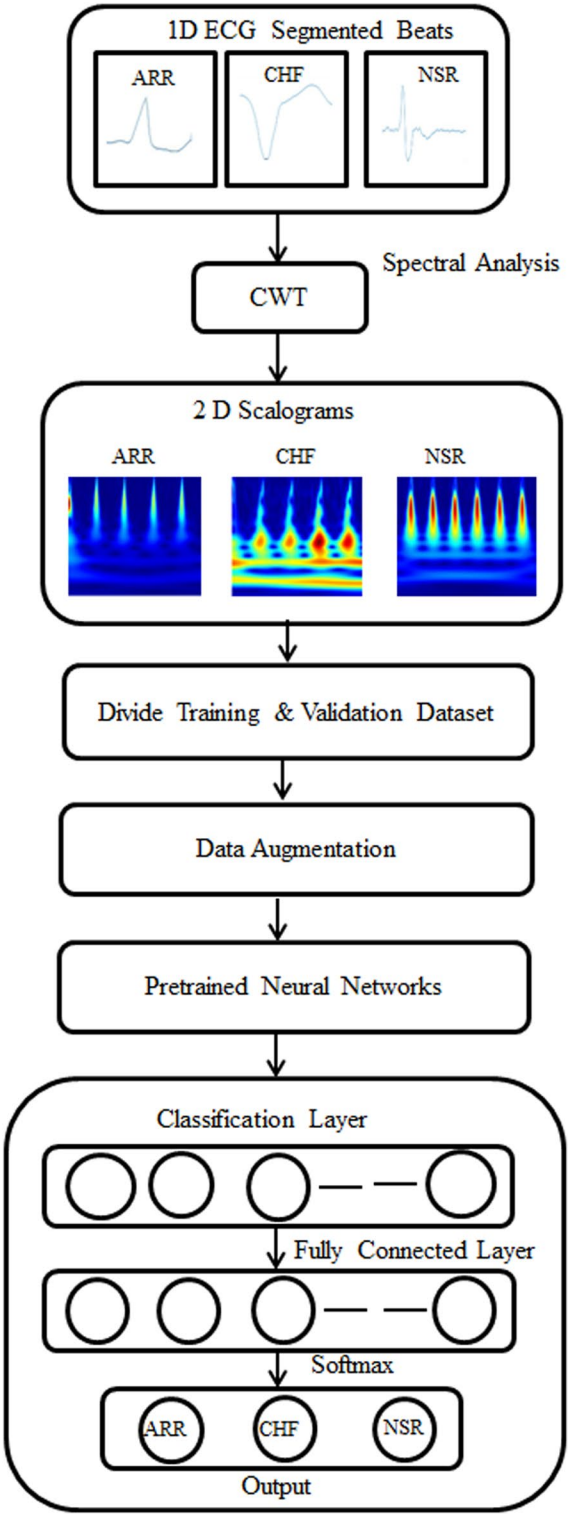


Fig. 1 Block diagram of our pre-trained CNN model

2.1 Proposed methodology

In this section, we first introduce the dataset. Next, we outline a transform for attaining 2D scalograms (images) from 1D ECG signals. Following that, we lead four training models intended for categorizing scalograms into three categories. Finally, we illustrate how it works of the suggested technique in Fig. 1.

2.1.1 Dataset

To determine the efficacy of the suggested technique, three separate ECG databases, ARR, NSR, and CHF, were employed for class labeling. The ARR class has 96 people, the CHF class has 30, and the NSR class has 36. All raw ECG records have been collected with a duration of 60 min and a 128-Hz sampling rate [26].

2.1.2 Continuous Wavelet Transform (CWT)

To analyze the signals in both the time and frequency domain, CWT is a great choice [15]. By using CWT, we can convert 1D signals into 2D scalograms, which give both spatial and temporal information. To acquire the modified signal, CWT convolves it with the wavelet functions of various scales with translation. CWT of a function f(t) is given by

$$\mathrm{CWT}(l,m) = \int_{-\infty}^{\infty} f(t)\psi_{l,m}\left(\frac{t-m}{l}\right)\mathrm{d}t. \tag{1}$$

where $\psi_{l,m}\left(\frac{t-m}{l}\right)$ indicates the shifted and scaled version of the mother wavelet function $\psi_{l,m}(t)$, and the shifted parameter is m & the scaling parameter is l. Equation (1) generates a matrix with wavelet coefficients at different scales and locations, and

Figure 2a,b, and c displays the conquered scalograms for three classes via CWT.

2.1.3 Pre-trained neural networks

DNNs are used for deploying classification models on resource-constrained devices like mobiles, especially for real-time signal analysis. Popular models, tailored for mobile compatibility, are evaluated based on accuracy and parameters. Employing transfer learning minimizes training time, with models initially trained on ImageNet and then adapted for ECG signal classification. Pre-trained networks used in this paper are AlexNet, Google Net, Vgg16, and SqueezeNet.

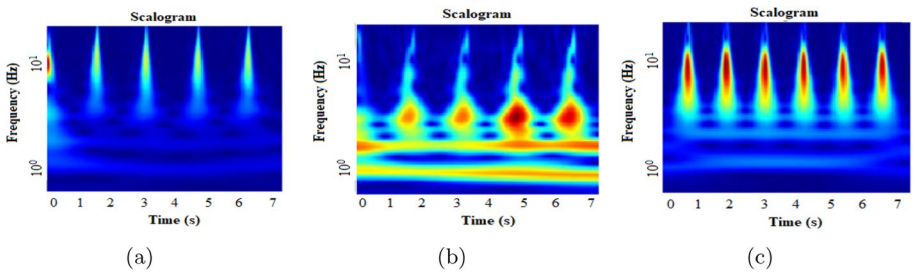


Fig. 2 2D scalograms of: **a** ARR, **b** CHF, **c** NSR

2.1.4 ECG classification

To achieve better classification accuracy, SqueezeNet is used as a pre-trained network and its block diagram and fire modules are displayed in Figs. 3 and 4. The fire module serves as the primary element of the network. In this network, the ReLU layer follows each convolutional layer. Conv1 is a convolutional layer with a 3x3 kernel and a stride of 2. For pooling, 3x3 max pooling layers with a stride of 2 are implemented.

SqueezeNet, an early compact model, achieves ImageNet-level accuracy like AlexNet with 50 times fewer parameters. It emphasizes small kernel sizes (1x1 and 3x3), notably employing fire modules that compress data via 1x1 convolution before expanding it with concatenated 1x1 and 3x3 convolutions. Originally comprising 1.2 million parameters across 68 layers, it is tailored for a 227 x 227 x 3 input image size, classifying into three categories. Modifications include adjusting dropout layers in the final six layers to 0.6, resulting in a current model with 724K parameters and 68 layers. The proposed CNN architecture processes input images through a series of convolutional, activation, pooling, and concatenation layers, culminating in classification outputs.

To compare the performance of the projected CNN algorithm, AlexNet, GoogleNet, Vgg16, and SqueezeNet are used. The deep learning models with CWT-AlexNet, CWT-GoogleNet, CWT-Vgg16, and CWT-SqueezeNet are referred to as TM1, TM2, TM3, and TM4, respectively.

3 Results and discussion

For the performance evaluation, the parameters accuracy (Acc), sensitivity (Se), specificity (Sp), Precision, F1- score, and Detection Error Rate (DER) are calculated by the mathematical equations from equation (2) to equation(7).

$$\text{Acc} = \frac{(\text{TP}'_1 + \text{TN}'_1)}{(\text{TP}'_1 + \text{TN}'_1) + (\text{FP}'_1 + \text{FN}'_1)}. \tag{2}$$

$$\text{Se} = \frac{(\text{TP}'_1)}{(\text{TP}'_1 + \text{FN}'_1)}. \tag{3}$$

$$\text{Sp} = \frac{(\text{TN}'_1)}{(\text{FP}'_1 + \text{TN}'_1)}. \tag{4}$$

$$\text{Precision} = \frac{(\text{TP}'_1)}{(\text{TP}'_1 + \text{FP}'_1)}. \tag{5}$$

$$\text{F1 Score} = \frac{2 * (\text{TP}'_1)}{2 * (\text{TP}'_1) + (\text{FP}'_1 + \text{FN}'_1)}. \tag{6}$$

$$\text{DER} = \frac{(\text{FP}'_1 + \text{FN}'_1)}{(\text{TP}'_1 + \text{TN}'_1) + (\text{FP}'_1 + \text{FN}'_1)}. \tag{7}$$

where TP'_1 = True positive beats FN'_1 = False negative beats TN'_1 = True negative beats FP'_1 = False positive beats

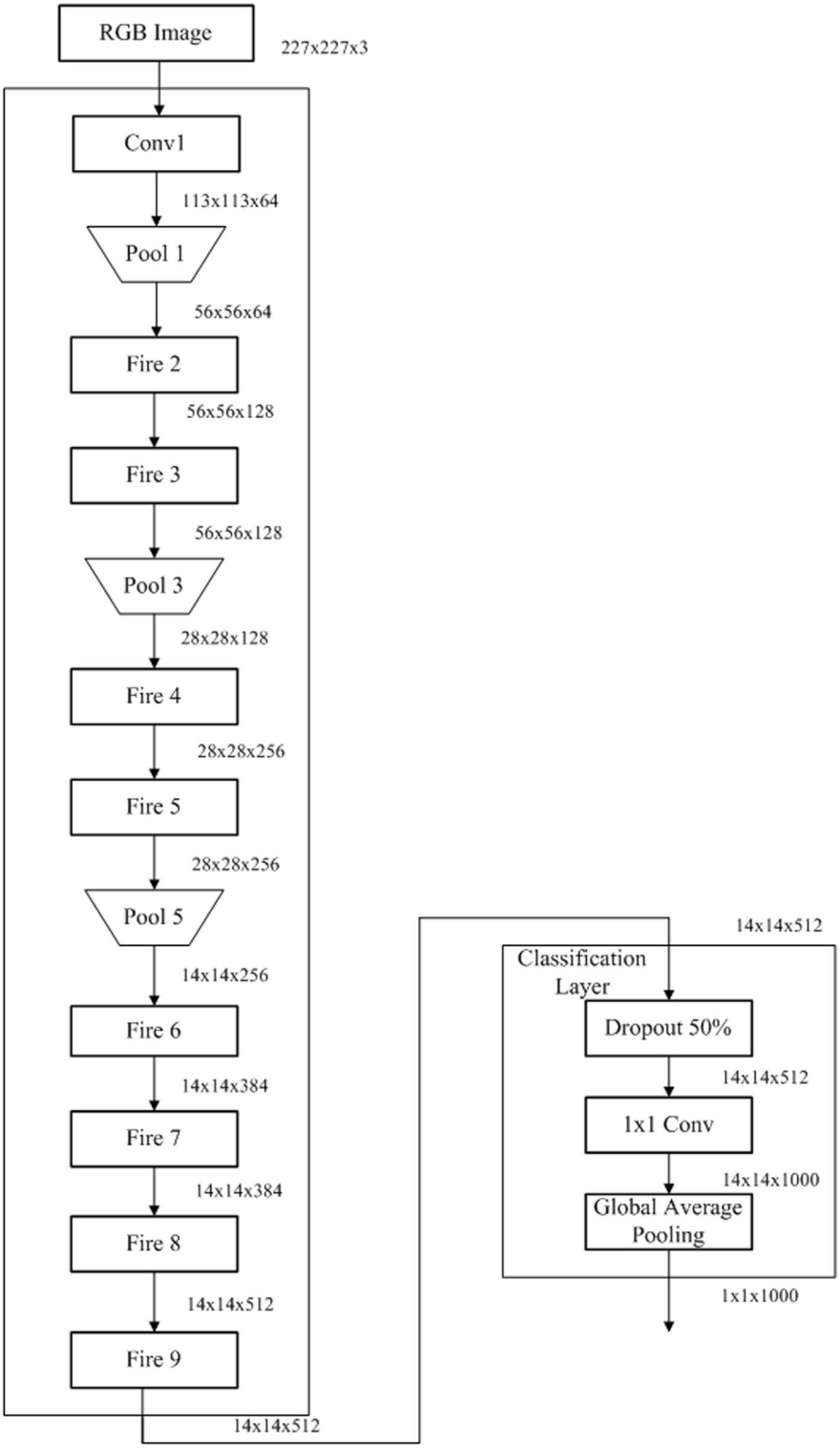


Fig. 3 Architecture of SqueezeNet

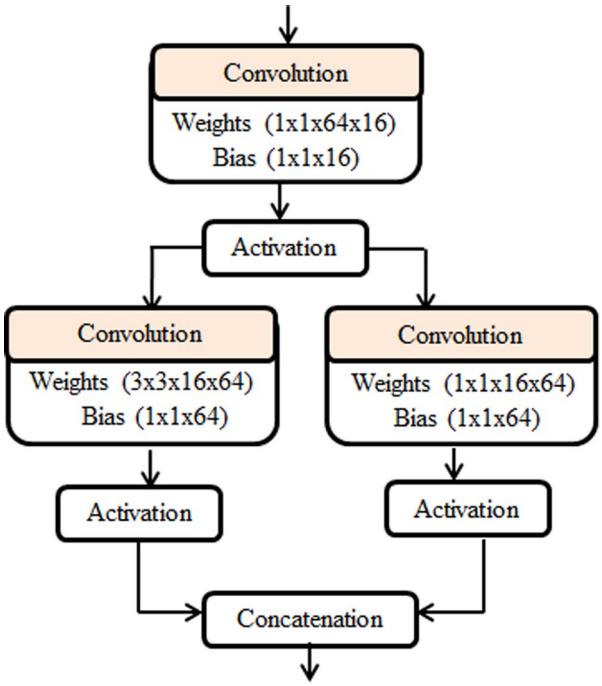


Fig. 4 Squeezenet fire module

Table 1 Hyperparameters and their values

Hyperparameters	Values
Learning rate	10^{-4}
Mini batch size	20
Max epochs	8
Validation frequency	10
Optimizer	Adaptive moment estimation

The hyperparameters used in the network are shown in Table 1, and the classification results in terms of various beats are presented in Table 2. Table 3 describes each class performance measures such as Acc, Se, Sp, F1-score, Precision, and DER for NSR, ARR, and CHF, respectively.

The average values of the CWT-SqueezeNet TM_4 model yield 98.7% of Acc, 99.1% of Se, 99.20% of Sp, 98.33% of F1-Score, 98.67% of Precision, and 0.89% of DER, which is clearly displayed in Table 4.

The proposed method demonstrated outstanding classification results across all classes and surpassed other methods in detecting arrhythmia. Consequently, the results accomplished in the current works are analogous to utmost allied works in deep learning experiments on a similar dataset. Bhati et al. [27] addressed a hybrid DL method combining CNN and BLSTM to categorize the MIT-BIH database, achieving a classification accuracy of 97.1%. In Kishore et al. [28], a pre-trained EfficientNet B0 CNN architecture was employed. To obtain 2D scalograms from 1D ECG records, STFT is applied and the

Table 2 Simulation results in terms of classification beats

Classifier	Classes	TP'_1	TN'_1	FP'_1	FN'_1
TM_1	ARR	46	98	4	2
	CHF	47	98	3	2
	NSR	50	51	0	3
TM_2	ARR	49	98	1	2
	CHF	49	99	3	1
	NSR	49	100	0	0
TM_3	ARR	49	97	1	3
	CHF	46	99	4	1
	NSR	50	98	0	1
TM_4	ARR	50	98	0	2
	CHF	50	100	0	0
	NSR	48	100	2	0

Table 3 Performance metrics of various models for individual classes

Classifier	Classes	Acc(%)	Se(%)	Sp(%)	F1-score(%)	Precision(%)	DER(%)
TM_1	ARR	95.83	96	96.1	93.88	92	4
	CHF	95.92	96.67	97	94.94	94	3.33
	NSR	94.34	67.33	100	97.08	100	2
TM_2	ARR	94.2	98	98.98	97.02	98	2.667
	CHF	97.9	98.67	96.11	96.07	92	3.33
	NSR	98	99.33	100	100	100	0.666
TM_3	ARR	96	97.33	98.99	96.07	98	2
	CHF	98	96.77	99	94.84	98	1.33
	NSR	100	99.33	99	99.01	98	0.667
TM_4	ARR	98.7	98.67	100	99.01	100	1.33
	CHF	100	100	100	98.03	100	0
	NSR	100	98.67	97.67	97.96	96	1.33

Table 4 Average values of the performance measures for various models

Proposed models	Acc(%)	Se(%)	Sp(%)	F1-score(%)	Precision(%)	DER(%)
TM_1	95.3	86.6	97.7	95.33	95.33	2.44
TM_2	96.7	98.7	98.67	97.69	96.67	2.22
TM_3	98.0	98.8	98.99	97.99	98	1.33
TM_4	98.7	99.1	99.20	98.33	98.67	0.89

achieved classification accuracy is 97.3% when testing the model on the standard PhysioNet database.

Karthiga et al. [29] implemented a deep CNN for ECG signal classification, reporting a classification accuracy of 91.92%. Similarly a hybrid CNN-LSTM was introduced [30] for detecting cardiac disorders of ECG records through CWT images. The authors [31, 32] developed a new classification mechanism with multistage Deep

Table 5 Summary of most allied mechanisms and present work

Method	Data Input	Model	Accuracy
Altan et al. [33]	1D ECG	DBN	97.3%
Strodthoff et al. [2]	1D ECG	ResNet and Inception	89.8%
Siddique et al. [3]	1D ECG	Deep extreme machine learning	92.4%
Hsu et al. [5]	1D ECG	AlexNet and ResNet	94.4%
Mewada et al. [25]	1D ECG	CNN layers	95.2%
Gaddam et al. [17]	2D ECG	AlexNet	95.6 %
Panda et al. [22]	1D ECG	CNN with EWT	97.6 %
Rahman et al. [20]	1D ECG	Alexnet	98.3 %
Ting et al. [13]	2D ECG	EfficientNet B0	97.1 %
Bhatia et al. [27]	1D ECG	LSTM+CNN	98.3 %
Karthiga et al. [29]	2D ECG	CNN	91.9 %
Madan et al. [30]	2D ECG	LSTM+CNN	97.3 %
Proposed TM_4	**2D ECG**	**Squeezenet**	**98.7** %

Bold values are used to differentiate proposed method results with used signal, model and obtained accuracy

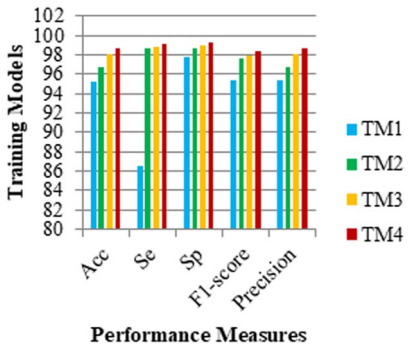


Fig. 5 Performance measures versus training models

Belief Network (DBN), and Altan et al. [33] produced an integrated method for the early detection of CHF. In this method, Hilbert Huang Transform (HHT) extracts the features and neural network classifies the data into five classes with an accuracy of 97.53% According to the stated results, the recommended method has 98.7% accuracy for ARR, 100% for CHF, and 100% for NSR. When compared to prior research that used other DL models on the same dataset, our model performed better in classifying ECG signals into ARR, CHF, and NSR classes. An overview of the most relevant studies carried out on the same database is provided in Table 5. The acquired outcomes are superior to the state-of -the-art methods and will considerably minimize the amount of contribution required by doctors.

Among all those four models TM4 gives an outstanding performance. Its average Acc is increased by 3.6%, 2.1%, and 0.7% to TM_1, TM_2, and TM_3 models. Se value is enhanced by 14.4%, 0.4%, and 0.3% than TM_1,TM_2, and TM_3 models, etc. But DER is reduced by 63.5%, 59.9%, and 3.5% than TM_1, TM_2, and TM_3 models. Finally, the entire model's performance analysis and their graphical representations are shown in Figs. 5 and 6. The corresponding confusion matrixes and their training progress for all models TM_1, TM_2, TM_3, and TM_4, are shown in Figs. 7, 8, 9, 10, and 11. Confusion

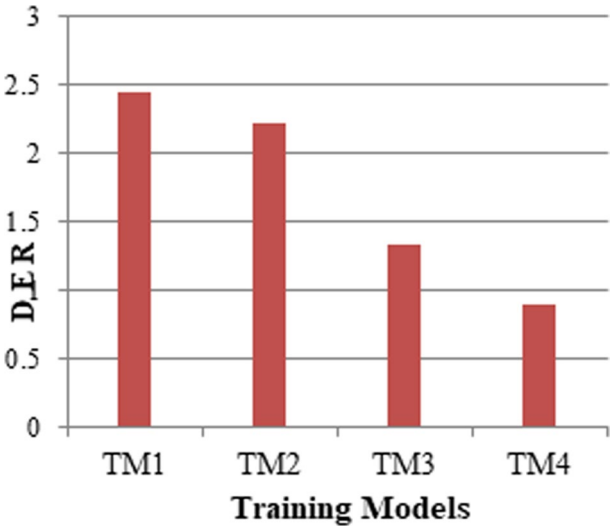


Fig. 6 Training models versus detection error rate

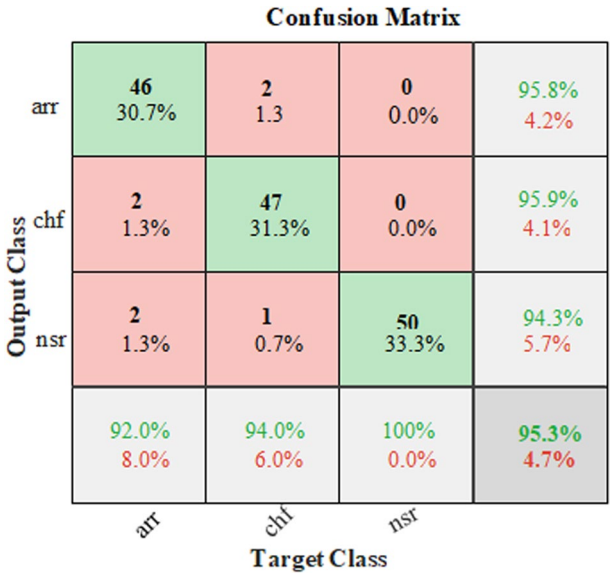


Fig. 7 Confusion matrix of TM1 model

matrices are important tools for visualizing and analyzing a model's predictions against the actual ground truth across several classes.

4 Conclusion

Arrhythmia classification is an incredibly important topic in healthcare. An arrhythmia is an erratic heartbeat or rhythm. This article presents a deep learning algorithm that uses alternative time-frequency representations of short ECG segments to reliably diagnose arrhythmia. CWT-RGB using a recommended 2D-CNN model has the best classification performance concerning average accuracy, sensitivity, specificity, F1-score, Precision, and DER at 98.7%, 99.1%, 99.2%, 98.3%, 98.7%, and 0.89%, respectively. The confusion matrix of the testing dataset exposed 96.2%, 100%, and 100%

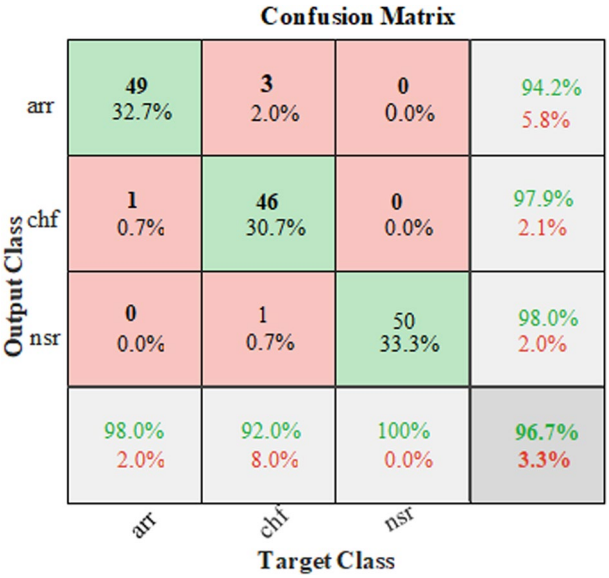


Fig. 8 Confusion matrix of TM2 model

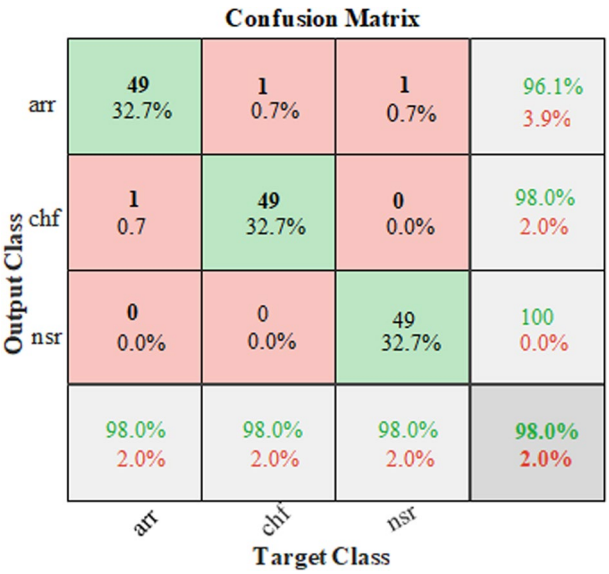


Fig. 9 Confusion matrix of TM3 model

validation accuracy for "ARR," "CHF," and "NSR." Furthermore, Se, Sp, F1-score, Precision, and DER for ARR are 98.67%, 100%, 99.01%, 100%, and 1.33% Se, Sp, F1-score, Precision, and DER for CHF are 100%, 100%, 98.03%, 100%, and 0%, and for NSR, Se, Sp, F1-score, Precision, and DER are 98.67% and 97.67%,97.96%,96%, and 1.33%, respectively. Similarly, the proposed methodology will help doctors accurately identify cardiac disorders during normal ECG monitoring. Furthermore, the high computing overhead imposed by the usage of CWT is a disadvantage. We could never attain total intersubjectivity, even if doing so would greatly reduce the scope of action

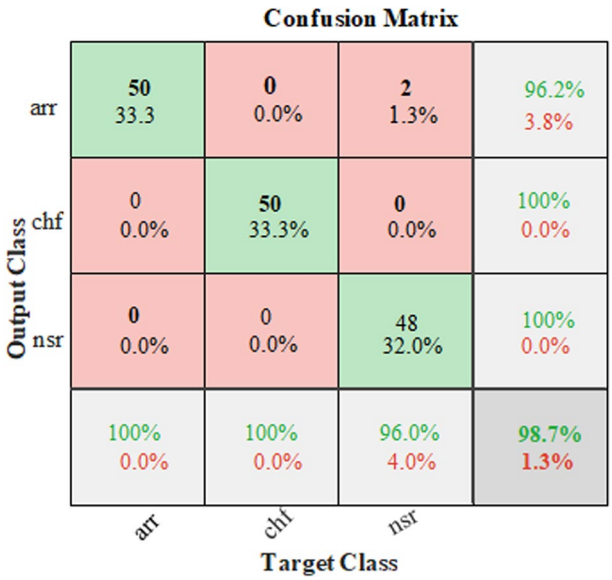


Fig. 10 Confusion matrix of TM4 model

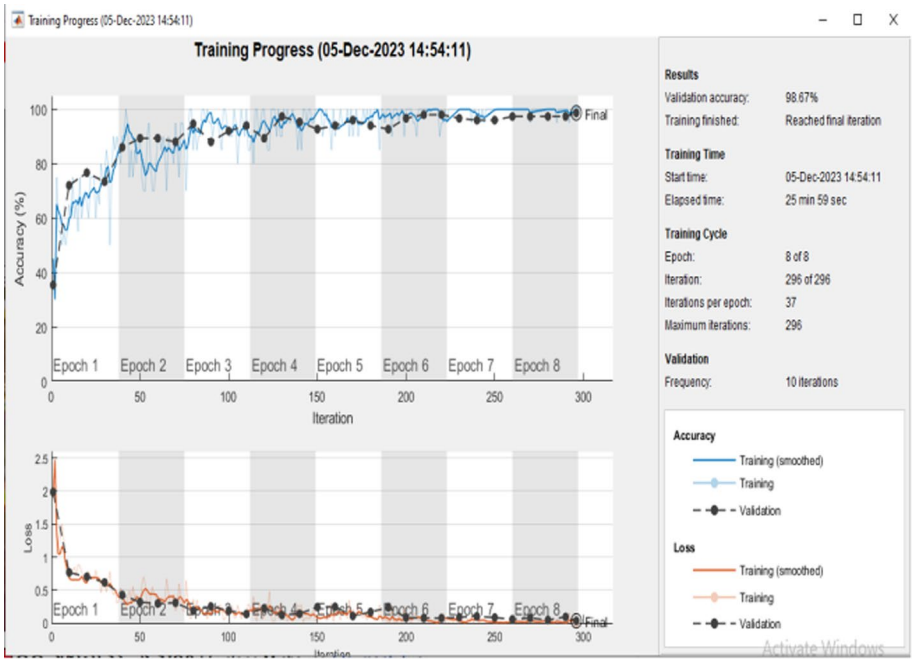


Fig. 11 Training progress of TM4 model

expected by physicians. It will be a fantastic upcoming opportunity for scholars. To deal with these issues, a strong arrhythmia classification framework is needed.

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All authors participated equally.

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